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Writing Philosophy

A Student's Guide to Writing Philosophy Essays

Lewis Vaughn

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2

How to Read an Argument

To a large extent, to read and write philosophy is to read and write arguments, for logical argument is at the heart of philosophy. When we do philosophy, we are usually either evaluating or constructing arguments. The philosophy that we read will most likely contain arguments, and our understanding of the text will hang on our ability to identify and understand those arguments. Most often the quality of the philosophical essays we write will depend heavily on the quality of the arguments we craft. Learning the fundamentals of logical argument, then, is a prerequisite for making sense out of philosophy and writing good philosophy papers. This chapter will get you going in the right direction by showing you how to pick out arguments from passages of nonargumentative prose, assess the quality of different kinds of arguments, devise good arguments, and recognize defective arguments when you see them.

PREMISES AND CONCLUSIONS

A *statement*, or claim, is an assertion that something is or is not the case. It is the kind of utterance that is *either true or false*. These are statements:

A tree is growing in the quad.

I am shocked and dismayed.

$2 + 2 = 4$

Time heals all wounds.

The universe is fifteen billion years old.

These, however, are *not* statements:

Why is a tree growing in the quad?

Why are you shocked and dismayed?

Stop being stupid.

Holy cow!

The first two sentences are questions; the third, a command or request; and the fourth, an exclamation. These are not the kind of things that can be true or false.

An *argument* is a combination of statements in which some of them are intended to support another one of them. That is, in an argument, some statements are intended to provide reasons for believing that another statement is true. The statements supposedly providing the support are known as *premises*; the statement being supported is known as the *conclusion*.

A basic task that rational beings engage in is trying to determine how strongly to believe a statement. A fundamental principle in philosophy (and our daily lives) is that the strength of our belief in a statement should depend on the strength of the reasons for believing it. A statement supported by strong reasons is worthy of strong acceptance. A statement supported by weaker reasons deserves weaker acceptance. The analysis of arguments is the primary method we use to discover how much credence to give statements and reasons. This method is essential not only in philosophy but in all fields of knowledge.

By now you probably have guessed that logical arguments are not arguments in the ordinary sense of shouting matches, heated debates, or angry squabbles. In philosophy and other kinds of intellectual exploration, an argument is a conclusion meant to be backed by reasons. Arguments in the ordinary sense are beside the point.

We must also distinguish between logical argument and persuasion. They are *not* synonymous. Presenting an argument is a way to demonstrate that a conclusion (a statement or belief) is warranted, that the conclusion is worthy of acceptance. This demonstration may or may not persuade someone to accept the conclusion; the argument's persuasiveness or lack thereof is a completely separate matter. On the other hand, you may be able to persuade someone to accept a statement through the use of psychological or rhetorical

trickery, emotional language, fallacious appeals, threats of force, deceit, and many other ploys. But if you do, you will not have shown that the statement is worthy of belief, for you will not have provided any reasons for accepting it.

Here are some arguments:

ARGUMENT 1

Chivalry is dead. My instructor says so.

ARGUMENT 2

If Michigan wins, there will be a riot in the stadium. They will definitely win. So the riot is a sure thing.

ARGUMENT 3

Ninety-eight percent of the students are conservatives. Joan is a student, so she's probably a conservative too.

ARGUMENT 4

All men are mortal. Socrates is a man. Therefore, Socrates is mortal.

Now let's label the premises and conclusion in each one:

ARGUMENT 1

[CONCLUSION] Chivalry is dead. [PREMISE] My instructor says so.

ARGUMENT 2

[PREMISE] If Michigan wins, there will be a riot in the stadium. [PREMISE] They will definitely win. [CONCLUSION] So the riot is a sure thing.

ARGUMENT 3

[PREMISE] Ninety-eight percent of the students are conservatives. [PREMISE] Joan is a student, [CONCLUSION] so she's probably a conservative too.

ARGUMENT 4

[PREMISE] All men are mortal. [PREMISE] Socrates is a man. [CONCLUSION] Therefore, Socrates is mortal.

Notice that these arguments differ in the placement and number of their parts. In argument 1, the conclusion comes first, then the premise. In the other three arguments, the conclusion comes last.

Argument 1 has one premise, but arguments 2, 3, and 4 have two premises each. The point is that arguments come in all kinds of configurations. An argument could have one premise, or two, or ten, or more. Sometimes a premise (or even the conclusion) is unstated, leaving the reader to fill in the blank (more on uncovering implicit premises later). But regardless of their structure, all arguments must have a conclusion and at least one premise. As long as this requirement is met, arguments can vary all over the map.

Now take a look at this passage:

> The stock market has tanked. Brokers are skittish. The Dow is the lowest it's been in ten years. We're pretty scared about all this.

Is there an argument here? If you think so, where is the conclusion? Where are the premises? Alas, this little observation about the stock market is not an argument. The passage consists of a series of statements with no conclusion in sight. There is no statement that other statements are supporting.

It is easy enough, however, to turn these statements into an argument. Look:

ARGUMENT 5

> The world of stocks and bonds is in for a rough ride because the stock market has tanked, brokers are skittish, and the Dow is the lowest it's been in ten years.

This passage is now a bone fide argument. The conclusion is "It's clear that the world of stocks and bonds is in for a rough ride." The three statements that follow support the conclusion.

Four Common Mistakes in Evaluating Arguments

1. Failing to distinguish between the logical structure of an argument and the truth of its premises.
2. Thinking that merely stating views is the same thing as presenting an argument.
3. Thinking that persuading someone to accept a claim is the same thing as presenting an argument.
4. Failing to distinguish between arguments and nonargumentative material.

Being able to distinguish arguments from nonargumentative material is a very valuable skill, and the sooner you master it the better. You will often encounter passages that seem to conceal an argument somewhere but do not. Many people (are you one of them?) think that if they clearly and firmly state their beliefs, they have given an argument. Sometimes they fill whole essays with many well-made statements and interesting points, thinking that they are making a strong case—but there is no argument anywhere. Despite all the verbiage, such essays fail to provide the reader with a single good reason to accept any assertion. As we have seen, a collection of unsupported statements does not an argument make. For there to be an argument, at least one statement must provide reasons for accepting another.

Now consider this passage:

> Johnson is absolutely the worst governor this state has ever had. I don't understand how anyone can read about the governor's exploits in the paper and still support him. Haven't the citizens of this state had enough scandal and corruption in the governor's office? I am aghast at Johnson's outrageous behavior and obvious ineptness.

Has the writer presented an argument? Once again, no. This passage is certainly an expression of displeasure and perhaps even of anger and disgust. But there is no claim supported with reasons. With some major alterations, however, we can turn the passage into an argument:

ARGUMENT 6

> Johnson is absolutely the worst governor this state has ever had. He has embezzled money from the state's treasury. He has ruined the state's economy. And he has used his office to persecute those he doesn't like. I am aghast at Johnson's outrageous behavior and obvious ineptness.

If we laid out the argument so we can more clearly see its structure (that is, if we outline it), it would look like this:

PREMISE 1: He has embezzled money from the state's treasury.

PREMISE 2: He has ruined the state's economy.

PREMISE 3: And he has used his office to persecute those he doesn't like.

CONCLUSION: Johnson is absolutely the worst governor this state has ever had.

Notice that the last sentence of the passage is not shown in this argument outline. That's because it is not part of the argument; it is not supporting the conclusion. It is simply an expression of the writer's reaction to the governor's misdeeds.

Logical arguments often come packaged with all sorts of nonargumentative material—introductory remarks, explanations, redundancies, descriptions, asides, examples, and more. The trick is to separate the premises and conclusion from all the other stuff. Once you pinpoint those, spotting the extraneous material is fairly straightforward.

As noted in Rule 1-3, the simplest way to locate an argument is to find its conclusion first, then its premises. Zeroing in on conclusions and premises can be a lot easier if you keep an eye out for *indicator words*. Indicator words often tag along with arguments and indicate that a conclusion or premise may be nearby. Arguments 2–5 have several indicator words. Arguments 2 and 3 contain the word *so*, which points to their respective conclusions. In argument 4 the indicator word *therefore* tells us that the conclusion follows. In argument 5 the indicator word *because* signals the presence of the three premises.

Here are a few conclusion indicator words:

consequently	as a result
thus	hence
therefore	so
it follows that	which means that

Here are some premise indicator words:

in view of the fact	assuming that
because	since
due to the fact that	for
the reason being	given that

Just remember that indicator words do not *guarantee* the presence of conclusions and premises. They are simply telltale signs.

There is one final wrinkle you should understand: Some arguments have unstated, or implicit, premises, and a few even have unstated conclusions. For example:

ARGUMENT 7

Any judge who supports capital punishment for juvenile offenders is an enemy of the Bill of Rights. Judge Simpson is definitely an enemy of the Bill of Rights.

The conclusion of this argument is "Judge Simpson is definitely an enemy of the Bill of Rights," and the only premise is the first statement. But there seems to be a leap of logic from the premise to the conclusion. Something is missing. The conclusion follows logically only if we insert an additional premise to bridge the gap, something like this:

ARGUMENT 7A

Any judge who supports capital punishment for juvenile offenders is an enemy of the Bill of Rights. Judge Simpson supports capital punishment for juvenile offenders. Therefore, Judge Simpson is definitely an enemy of the Bill of Rights.

Now we can see both the stated premise as well as the one left unmentioned, and we can conduct a complete evaluation.

When you're evaluating an argument, you should always bring any implicit premises out into the open. Finding the implicit premise makes evaluation easier. It also helps you avoid falling into a logical trap. Often premises that are left unstated are dubious or false. You should never let questionable premises slide by unnoticed.

JUDGING ARGUMENTS

In Chapter 1, we saw the importance of approaching a philosophical text with an open mind (Rule 1-1); reading actively and critically (Rule 1-2); identifying the conclusion first, then the premises (Rule 1-3); outlining, paraphrasing, or summarizing the argument (Rule 1-4); and evaluating the argument and formulating a tentative judgment (Rule 1-5). Chances are, you are most intimidated by Rule 1-5. You need not be. The following rules elaborate on Rule 1-5 and detail more techniques for systematically sizing up arguments, even complex ones that are buried in a tangle of extraneous material.

Rule 2-1 Know the Basics of Deductive and Inductive Arguments

As we saw in Chapter 1, a good argument gives us good reasons for accepting a conclusion; a bad argument fails to provide good reasons. To tell the difference—and to do so consistently—you need to understand the different forms that arguments can take.

Arguments come in two basic types: *deductive* and *inductive*. Deductive arguments are supposed to offer logically *conclusive* support for their conclusions. If a deductive argument actually manages to provide logically conclusive support for its conclusion, it is said to be *valid*. If it fails to provide logically conclusive support for the conclusion, it is said to be *invalid*. If a deductive argument is valid, it has this peculiar characteristic: If its premises are true, its conclusion absolutely *must be true*. In a valid argument, *it is impossible for the premises to be true and the conclusion false*. Notice that *valid* is not a synonym for *true*. A valid argument is simply one that has the kind of structure, or form, that *guarantees* that if the premises are true, the conclusion is also true. A valid argument, then, is one whose conclusion *follows from the premises*.

Take a look at this simple deductive argument:

ARGUMENT 8

All soldiers are brave.

Roy is a soldier.

Therefore, Roy is brave.

In this argument, you can see that if the premises are true, then the conclusion absolutely must be true. There is no way that the conclusion can be anything else but true. The form of the argument—not its content—is what guarantees this outcome. Notice that the first premise is actually false, but the argument is still valid. We could plug completely different statements into this argument, but as long as the form stays the same, the argument would remain valid.

Consider this deductive argument:

ARGUMENT 9

If stealing harms people, then it is morally wrong.

Stealing does harm people.

Therefore, stealing is morally wrong.

This argument is also valid. If it is true that *if stealing harms people then it is morally wrong*, and if it is also true that *stealing does harm people*, then the conclusion (*stealing is morally wrong*) must be true as well.

Traditionally, philosophers have symbolized the form of deductive arguments by using letters to stand for parts of the argument. The form of argument 9, then, can be symbolized like this:

If p, then q.

p.

Therefore, q.

Here, the letters p and q represent the two statements in the argument. Notice that the first premise is a compound statement, which is made up of two constituent statements p and q. We could insert any statements we want into this valid form, and it would still be the case that if the premises are true, the conclusion must be true.

Unlike deductive arguments, inductive arguments are meant to provide *probable* support for their conclusions. If an inductive argument succeeds in providing such probable support, it is said to be *strong*. In a strong argument, if the premises are true, the conclusion is probably true. If an inductive argument fails to provide probable support for its conclusion, it is said to be *weak*. Like a valid argument, a strong argument is one whose conclusion *follows from the premises*.

So inductive arguments cannot guarantee the truth of their conclusions as deductive arguments can. They can only render their conclusions probable—that is, more likely to be true than not. In inductive arguments, then, it is possible for the premises to be true and the conclusions false.

Consider these two inductive arguments:

ARGUMENT 10

Almost all of the students at this school are Democrats.

Therefore, Maria, who is a student here, is probably a Democrat too.

ARGUMENT 11

Ninety percent of the Republicans I know are Volvo owners.

Therefore, 90 percent of all Republicans are probably Volvo owners.

Argument 10 is strong. If it is true that almost all the students at the school are Democrats, then it is likely that Maria is a Democrat too. Yet it is possible for the premise of argument 10 to be true and the conclusion false.

Argument 11, on the other hand, is weak. Even if 90 percent of the Republicans one person knows own Volvos, it does not follow that *all* Republicans are Volvo owners. The relatively small sample of Republican Volvo owners does not allow us to generalize to millions of Republicans.

We can now be more precise about the characteristics of good arguments. Good arguments (whether deductive or inductive) must be well reasoned—that is, they must be either valid or strong. But good arguments must also have true premises. An argument is not good (does not give us good reasons for accepting a conclusion) unless it is valid or strong *and* all of its premises are true. A valid deductive argument with true premises is said to be *sound*. A strong inductive argument with true premises is said to be *cogent*. So a good deductive argument is sound; a good inductive argument is cogent.

Rule 2-2 Determine Whether the Conclusion Follows from the Premises

Usually, the first step in assessing the worth of an argument is to determine whether the conclusion follows from the premises—that is, whether the argument is valid or strong. If the conclusion does not follow from the premises, then the argument cannot give you good reasons to accept the conclusion. The argument is bad—even if the premises are true.

Very often, when you examine an argument, you will see right away whether the conclusion follows from the premises. At other times, you may have to think about the argument's structure for a while. But in many cases, you will need some help in sizing up the argument—the kind of help offered in the following pages.

Fortunately, deductive arguments often occur in certain classic patterns. The forms, or structures, of the arguments show up again and again. Being familiar with these patterns can help you quickly determine whether an argument is valid.

Consider argument 9:

ARGUMENT 9

If stealing harms people, then it is morally wrong.

Stealing does harm people.

Therefore, stealing is morally wrong.

We symbolized this argument like this:

If p, then q.

p.

Therefore, q.

This kind of argument is known as a *conditional* (also, *hypothetical*). A conditional argument contains at least one conditional, or if-then, premise (If p, then q). The first half of a conditional premise (the *if* part) is called the *antecedent*. The second half (the *then* part) is known as the *consequent*. It happens that argument 9 is a conditional argument in a classic pattern called *modus ponens* (or *affirming the antecedent*). In *modus ponens*, the second premise affirms the antecedent of the first premise. Any argument using this pattern or form is *always* valid. So if you come across an argument in this pattern, you will know that it is valid—no matter what the statements say.

Another classic conditional pattern is called *modus tollens*, or denying the consequent. Look:

ARGUMENT 12

If the cat is on the mat, then she is asleep.

But she is not asleep.

Therefore, she is not on the mat.

ARGUMENT 13

If the mind is identical to the brain, then damaging the brain will damage the mind.

But damaging the brain will not damage the mind.

Therefore, the mind is not identical to the brain.

Modus tollens, then, is symbolized like this:

If p, then q.

Not q.

Therefore, not p.

Any argument having this form is valid.

Here's a slightly more involved conditional form known as *hypothetical syllogism:*

ARGUMENT 14

If the cat is on the mat, then she is asleep.

If she is asleep, then she is dreaming.

Therefore, if the cat is on the mat, then she is dreaming.

The hypothetical syllogism is symbolized like this:

If p, then q.

If q, then r.

Therefore, if p, then r.

Any argument having this form is valid.

QUICK REVIEW: Valid Conditional Argument Forms

AFFIRMING THE ANTECEDENT (*MODUS PONENS*)	DENYING THE CONSEQUENT (*MODUS TOLLENS*)
If p, then q.	If p, then q.
p.	Not q.
Therefore, q.	Therefore, not p.

HYPOTHETICAL SYLLOGISM

If p, then q.

If q, then r.

Therefore, if p, then r.

There are also common forms that are *not* valid. This one is known as *denying the antecedent:*

ARGUMENT 15

If the cat is on the mat, then she is asleep.

She is not on the mat.

Therefore, she is not asleep.

Denying the antecedent is symbolized like this:

If p, then q.

Not p.

Therefore, not q.

Finally, there is this invalid form, called *affirming the consequent.*

ARGUMENT 16

If the cat is on the mat, then she is asleep.

She is asleep.

Therefore, she is on the mat.

Affirming the consequent is symbolized like this:

If p, then q.

q.

Therefore, p.

The best way to use these argument forms to evaluate deductive arguments is to memorize them so you can more easily identify examples of the forms when you encounter them. Then you need only match the form of the deductive argument you are evaluating with one of these classic forms. If the argument matches one of the valid forms, it is valid; if one of the invalid forms, it is invalid.

QUICK REVIEW: Invalid Conditional Argument Forms

DENYING THE ANTECEDENT	AFFIRMING THE CONSEQUENT
If p, then q.	If p, then q.
Not p.	q.
Therefore, not q.	Therefore, p.

Inductive arguments also have distinctive forms, and being familiar with the forms can help you evaluate the arguments. Let's look at three common forms of inductive arguments.

In *enumerative induction*, we arrive at a generalization about an entire group of things after observing just some members of the group. Here is a typical enumerative inductive argument:

ARGUMENT 17

Every formatted disk I have bought from the computer store is defective.

Therefore, all formatted disks sold at the computer store are probably defective.

ARGUMENT 18

All the hawks in this wildlife sanctuary that I have observed have had red tails.

Therefore, all the hawks in this sanctuary probably have red tails.

ARGUMENT 19

Sixty percent of the Bostonians I have interviewed in various parts of the city are pro-choice.

Therefore, 60 percent of all Bostonians are probably pro-choice.

As you can see, enumerative induction has this form:

X percent of the observed members of group *A* have property *P*.

Therefore, X percent of all members of group *A* probably have property *P*.

The observed members of the group are simply a sample of the entire group. So based on what we know about this sample, we can generalize to all the members. But how do we know whether such an argument is strong? Everything depends on the sample. If the sample is large enough and representative enough, we can safely assume that our generalization drawn from the sample is probably an accurate reflection of the whole group of members. A sample is representative of an entire group only if each member of the group has an equal chance of being included in the sample. In general the larger the sample, the greater the probability that it accurately reflects the nature of the group as a whole. Often common sense tells us when a sample is too small.

We do not know how many formatted disks from the computer store are in the sample mentioned in argument 17. But if the number is several dozen and the disks were bought over a period of weeks or months, the sample is probably sufficiently large and representative. If so, the argument is strong. Likewise, in argument 18 we don't know the size of the sample or how it was obtained. But if the sample was taken from all the likely spots in the sanctuary where hawks live, and if several hawks were observed in each location, the sample is probably adequate—and the argument is strong. In argument 19, if the sample consists of a handful of Bostonians interviewed on a few street corners, the sample is definitely inadequate and the argument is weak. But if the sample consists of several hundred people, and if

every member of the whole group has an equal chance of being included in the sample, then the sample would be good enough to allow us to accurately generalize about the whole population. Typically, selecting such a sample of a large population is done by professional polling organizations.

In the argument form known as *analogical induction* (or *argument by analogy*), we reason in this fashion: Two or more things are similar in several ways; therefore, they are probably similar in one further way. Consider this argument:

ARGUMENT 20

Humans can walk upright, use simple tools, learn new skills, and devise deductive arguments.

Chimpanzees can walk upright, use simple tools, and learn new skills.

Therefore, chimpanzees can probably devise deductive arguments.

This argument says that because chimpanzees are similar to humans in several respects, they probably are similar to humans in one further respect.

Here's an argument by analogy that has become a classic in philosophy:

ARGUMENT 21

A watch is a complex mechanism with many parts that seem arranged to achieve a specific purpose—a purpose chosen by the watch's designer. In similar fashion, the universe is a complex mechanism with many parts that seem arranged to achieve a specific purpose. Therefore, the universe must also have a designer.

We can represent the form of an argument by analogy in this way:

X has properties P_1, P_2, P_3 plus the property P_4.

Y has properties P_1, P_2, and P_3.

Therefore, *Y* probably has property P_4.

The strength of an analogical induction depends on the relevant similarities between the two things compared. The more relevant similarities there are, the greater the probability that the conclusion is true. In argument 20, several similarities are noted. But there are

some unmentioned dissimilarities. The brain of a chimpanzee is smaller and more primitive than that of a human, a difference that probably inhibits higher intellectual functions such as logical argument. Argument 19, then, is weak. A common response to argument 21 is that the argument is weak because, although the universe resembles a watch in some ways, in other ways it does not resemble a watch. Specifically, the universe also resembles a living thing.

The third type of inductive argument is known as *inference to the best explanation*, a kind of reasoning that we all use daily and that is at the heart of scientific investigations. Recall that an argument gives us reasons for believing *that* something is the case. An *explanation*, on the other hand, states *how* or *why* something is the case. It attempts to clarify or elucidate, not offer proof. For example:

1. Megan definitely understood the material, for she could answer every question on the test.
2. Megan understood the material because she has a good memory.

Sentence 1 is an argument. The conclusion is "Megan definitely understood the material," and the reason (premise) given for believing that the conclusion is true is "for she could answer every question on the test." Sentence 2, however, is an explanation. It does not try to present reasons for believing something; it has nothing to prove. Instead, it tries to show why something is the way it is (why Megan understood the material). Sentence 2 assumes that Megan understood the material and then tries to explain why. Such explanations play a crucial role in inference to the best explanation.

In this type of inductive argument, we begin with premises about a phenomenon or state of affairs to be explained. Then we reason from those premises to an explanation for that state of affairs. We try to produce not just any old explanation but the best explanation among several possibilities. The best explanation is the one most likely to be true. The conclusion of the argument is that the preferred explanation is indeed probably true. For example:

ARGUMENT 22

Tariq flunked his philosophy course. The best explanation for his failure is that he didn't read the material. Therefore, he probably didn't read the material.

ARGUMENT 23

Ladies and gentlemen of the jury, the defendant was found with the murder weapon in his hand, blood on his clothes, and the victim's wallet in his pocket. We have an eyewitness putting the defendant at the scene of the crime. The best explanation for all these facts is that the defendant committed the murder. There can be very little doubt—he's guilty.

Here's the form of inference to the best explanation:

Phenomenon *Q*.

E provides the best explanation for *Q*.

Therefore, it is probable that *E* is true.

In any argument of this pattern, if the explanation given is really the best, then the argument is inductively strong. If the explanation is not the best, the argument is inductively weak. If the premises of the strong argument are true, then the argument is cogent. If the argument is cogent, then we have good reason to believe that the conclusion is true.

The biggest challenge in using inference to the best explanation is determining which explanation is the best. Sometimes this feat is easy. If our car has a flat tire, we may quickly uncover the best explanation for such a state of affairs. If we see a nail sticking out of the flat, and there is no obvious evidence of tampering or of any other extraordinary cause (that is, there are no good alternative explanations), we may safely conclude that the best explanation is that a nail punctured the tire.

In more complicated situations, we may need to do what scientists do to evaluate explanations, or theories—use special criteria to sort through the possibilities. Scientists call these standards the criteria of adequacy. Despite this fancy name, these criteria are basically just common sense, standards that you have probably used yourself.

One of these criteria is called *conservatism*. This criterion says that, all things being equal, the best explanation or theory is the one that fits best with what is already known or established. For example, if a friend of yours says—in all seriousness—that she can fly to the moon without using any kind of rocket or spaceship, you probably wouldn't believe her (and might even think that she needed psychiatric help). Your reasons for doubting her would probably rest on

the criterion of conservatism—that what she says conflicts with everything science knows about space flight, human anatomy, aerodynamics, laws of nature, and much more. It is logically possible that she really can fly to the moon, but her claim's lack of conservatism (the fact that it conflicts with so much of what we already know about the world) casts serious doubt on it.

Here is another useful criteria for judging the worth of explanations: *simplicity*. Other things being equal, the best explanation is the one that is the simplest—that is, the one that rests on the fewest assumptions. The theory making the fewest assumptions is less likely to be false because there are fewer ways for it to go wrong. In the example about the flat tire, one possible (but strange) explanation is that space aliens punctured the tire. You probably wouldn't put much credence in this explanation because you would have to assume too many unknown entities and processes—namely, space aliens who have come from who knows where using who knows what methods to move about and puncture your tires. The nail-in-the-tire theory is much simpler (it assumes no unknown entities or processes) and therefore much more likely to be true.

Rule 2-3 Determine Whether the Premises Are True

When you are carefully reading an argument (whether in an essay or some other context), you will be just as interested in whether the premises are true as in whether the conclusion follows from the premises. If the writer is conscientious, he or she will try to ensure that each premise is either well supported or in no need of support (because the premise is obvious or agreed to by all parties). The needed support will come from the citing of examples, statistics, research, expert opinion, and other kinds of evidence or reasons. This arrangement means that each premise of the primary argument may be a conclusion supported in turn by premises citing evidence or reasons. In any case, you as the reader will have to carefully evaluate the truth of all premises and the support behind them.

When you are trying to write a good argument, the story will be much the same. You will want to provide good reasons to your readers for accepting the premises, for you understand that simply explaining your premises is not enough. You will have to provide support for each premise requiring it and ensure that the support is adequate and reliable. (See Chapter 6 for guidance on using and citing sources.)

APPLYING THE RULES

Let's read and evaluate an argument in an extended passage, applying the rules discussed in this chapter (as well as those in the last). The following excerpt focuses on one aspect of a famous argument for the existence of God. Read it and review the comments that follow.

The Design Argument
Nigel Warburton

1. One of the most frequently used arguments for God's existence is the Design Argument, sometimes also known as the Teleological Argument (from the Greek word '*telos*' which means 'purpose'). This states that if we look around us at the natural world we can't help noticing how everything in it is suited to the function it performs: everything bears evidence of having been designed. This is supposed to demonstrate the existence of [a] Creator. If, for example, we examine the human eye, we see how its minute parts all fit together, each part cleverly suited to what it was apparently made for: seeing. . . .

2. Even if, despite the objections mentioned so far, you still find the Design Argument convincing, you should notice that it doesn't prove the existence of a unique, all-powerful, all-knowing, and all-good God. Close examination of the argument shows it to be limited in a number of ways.

3. First, the argument completely fails to support monotheism—the view that there is just one God. Even if you accept that the world and everything in it clearly shows evidence of having been designed, there is no reason to believe that it was all designed by one God. Why couldn't it have been designed by a team of lesser gods working together? After all, most large-scale, complex human constructions such as skyscrapers, pyramids, space rockets, and so on, were made by teams of individuals, so surely if we carry the analogy to its logical conclusion it will lead us to believe that the world was designed by a group of gods.

4. Second, the argument doesn't necessarily support the view that the Designer (or designers) was all-powerful. It could plausibly be argued that the universe has a number of 'design faults': for instance, the human eye has a tendency to short-sightedness and to cataracts in old age—hardly the work of an all-powerful Creator wanting to create the best world possible. Such observations might lead some people to think that the Designer of the universe, far from being all-powerful, was a comparatively weak God or gods, or possibly a young god experimenting with his or her powers. Maybe the Designer died soon after creating the universe, allowing it to run down of its own accord. The Design Argument provides at least as much evidence for these conclusions as it does for the

existence of the God described by the Theists. So the Design Argument alone cannot prove that the Theists' God rather than some other type of God or gods exists.[1]

To be successful in evaluating the argument, we must approach the text with an open mind (Rule 1-1) and read actively and critically (Rule 1-2), striving for full understanding. After that, our top priority is to identify the conclusion and the premises (Rule 1-3) and then outline, paraphrase, or summarize the whole argument (Rule 1-4). Our ultimate task is to evaluate the argument and formulate a tentative judgment about it (Rule 1-5). And that means having a good understanding of different types of arguments (Rule 2-1) so we can determine whether the argument is valid or strong (Rule 2-2) and whether the premises are true (Rule 2-3).

After reading the excerpt a few times, we can see that the author's purpose is to present an argument against another argument, the so-called design argument. In paragraph 1, he explains that the design argument says that because everything in the world looks as though it has been designed, it must have a designer, namely God. But in paragraph 2, he asserts that the design argument doesn't prove what many people think it does. It doesn't prove that the designer is God in the traditional sense—an all-powerful, all-knowing, all-good supreme being. He backs up this assertion in paragraphs 3 and 4, giving two reasons for believing that the creator doesn't necessarily have to be anything like the traditional God.

After reading the excerpt, we should go back and look for indicator words. When we do, we find only one—the conclusion indicator word *So* in the last sentence of paragraph 4. The statement that *So* introduces is indeed the conclusion of the argument: "So the Design Argument alone cannot prove that the Theists' God [the traditional deity] rather than some other type of God or gods exists." This sentence, however, is a reiteration of the conclusion, which is also stated (in different words) for the first time in paragraph 2: "[The design argument] doesn't prove the existence of a unique, all-powerful, all-knowing, and all-good God."

After identifying the conclusion, we can see that the first premise must be in paragraph 3: "Even if you accept that the world and everything in it clearly shows evidence of having been designed, there is

[1]Excerpted from Chapter 1 of Nigel Warburton, *Philosophy: The Basics* (London: Routledge, 2000), 12, 14–15. Reproduced by permission of the publisher.

no reason to believe that it was all designed by one God." Perhaps the world was designed by many gods. The design argument gives us no reason to rule out this possibility.

The second premise, then, is in paragraph 4: "the [design] argument doesn't necessarily support the view that the Designer (or designers) was all-powerful." Perhaps the designer was weak, or inexperienced, or mortal. The design argument doesn't prove otherwise.

When we outline the argument, we get this:

PREMISE 1: Even if you accept that the world and everything in it clearly shows evidence of having been designed, there is no reason to believe that it was all designed by one God.

PREMISE 2: The [design] argument doesn't necessarily support the view that the Designer (or designers) was all-powerful.

CONCLUSION: So the design argument alone cannot prove that the Theists' God rather than some other type of God or gods exist.

For the sake of clarity, we can paraphrase the argument:

PREMISE 1: The design argument doesn't show that the world must have been created by just one God.

PREMISE 2: The design argument doesn't show that world must have been created by an all-powerful God.

CONCLUSION: Therefore, the design argument doesn't show that the traditional God (all-powerful, all-knowing, and all-good) exists rather some other type of God or gods.

Now we ask the crucial question mandated by Rule 2-2: Does the conclusion follow from the premises? In this case, we would have to say yes. If the design argument fails to show that the creator is one God, and if it fails to show that the creator is all-powerful, then we must conclude that the design argument fails to show that the traditional God exists. That is, the author's argument is valid.

Next we ask the question prompted by Rule 2-3: Are the premises true? By studying the excerpt, we can see that the design argument, if strong, establishes only that the world probably had a designer (or designers). The argument does not show, however, that the designer had any distinguishing characteristics, including attributes associated with the traditional notion of God. If this is the case, then premise 1 must be true—the designer could have been one or many. Likewise, premise 2 must also be true—the designer could have been less than all-powerful.

Based on this analysis, we can say that the author's argument is sound. The design argument—in the form presented in our excerpt—does not prove the existence of a God with traditional attributes. Our analysis, however, is not the final word on the topic. There are other arguments relevant to the issues raised that we have not considered.

QUICK REVIEW: Basic Definitions

- *Statement:* An assertion that something is or is not the case. A statement is either true or false.
- *Argument:* A combination of statements in which some of them are intended to support another of them. Statements supposedly providing the support are the *premises*; the statement being supported is the *conclusion*.
- *Deductive argument:* An argument that is supposed to offer logically conclusive support for its conclusion. Deductive arguments can be *valid* or *invalid*. A valid argument with true premises is said to be *sound*.
- *Inductive argument:* An argument meant to offer probable support for its conclusion. Inductive arguments can be *strong* or *weak*. A strong argument with true premises is said to be *cogent*.